Implementation of Dielectric ELC in ESPResSo

Bachelor's Thesis

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Simulation Technology B.Sc.

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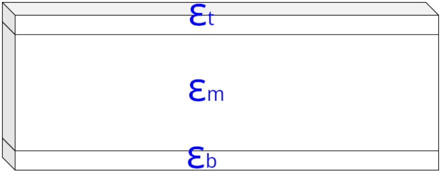
Motivation

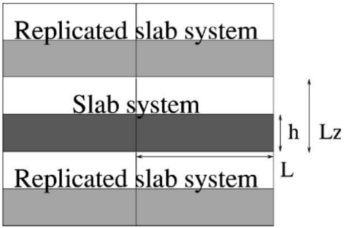
There already is an ELC implementation that can handle dielectric interfaces in ESPResSo, but:

- It has errors that are poorly understood.
- The code is unmaintainable and very difficult to read.

Goal: Replace the old ELC implementation with a clean, fixed, modern version.

Introduction





Developing in Python allows for faster iteration:

- Easier to develop, test, and prototype.
- Simplified visualization and validation of results.

Regular ELC: Development Strategy

Iterative/test-driven development by breaking the problem down:

- **Workflow per system:**
  - Compute reference solutions (analytical or brute-force).
  - Adjust ELC prototype to match reference solutions.
- **Systems analyzed:**
  - Large boxes (negligible PBC influence with $L_x, L_y >> 1$).
  - Smaller, neutral boxes (no non-neutral correction term yet).
  - Full parameter sweeps (Non-neutral, Madelung, random, and extreme parameters).

$$E_{2D+h} = E_{3D} + E_{\text{corr}} + E_{\text{non-neutral}} + E_{\text{dipole}}$$ [1]

[1] Tyagi, S., Arnold, A., & Holm, C. (2008). J. Chem. Phys. 129, 204102.

Regular ELC: Results

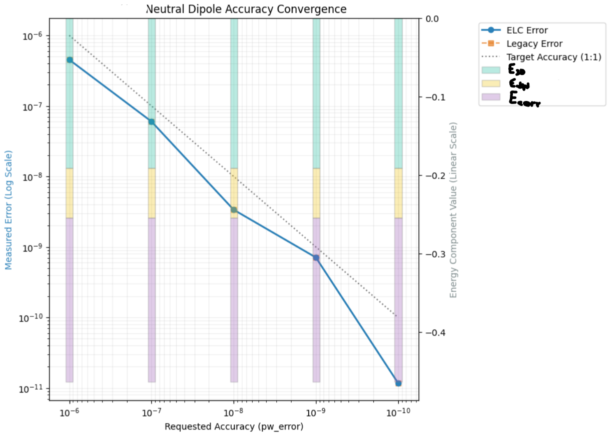


Figure: Validation of Regular ELC

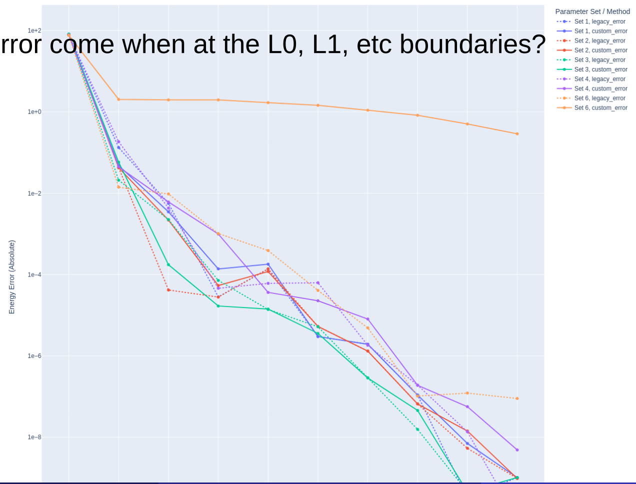
ELC Force Calculation: derived directly from the energy calculation $\vec{F} = -\nabla \vec{E}$

Dielectric ELC: Single Plate (Current Status)

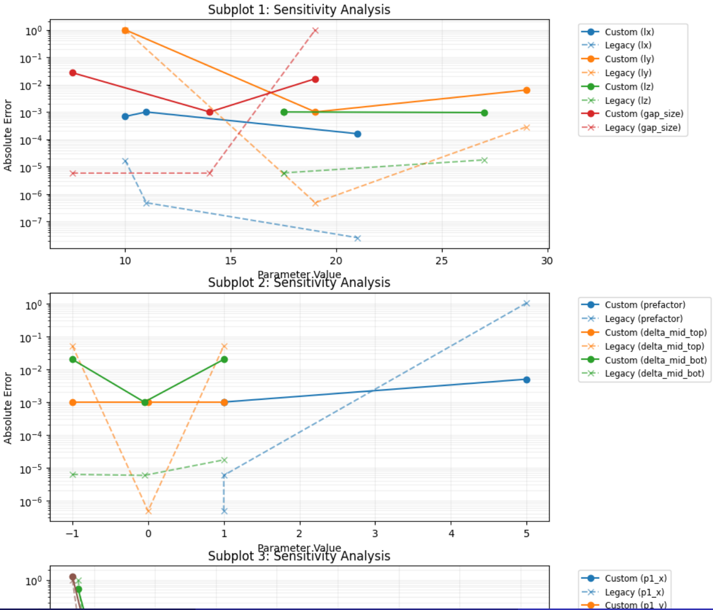
- Reference used: Ewald sum.
- Works for most parameters, though some outliers remain.

P3m doesnt know about my plate. my Ecorr, etc need to cancel it out

Does the Error come when at the L0, L1, etc boundaries?

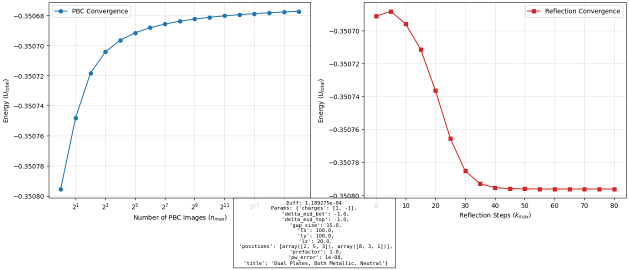


Dielectric ELC: Neighbourhood Scan



Dielectric ELC: Dual Plates

- Reference: Brute Force summation failed to converge.



there may be
water and sol method

Implementing in ESPResSo

- Upcoming: C++ implementation.
- Integration with **Kokkos** support for hardware acceleration.